Gpap — backup slides

detail behind the main deck

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Backup: the engine, in features

- one packed `int32` per clause — update is a single atomic
- assignment as a bitmap — one atomic gives dedup *and* contradiction
- two propagate kernels, **auto-dispatched** per formula
- the whole decide+propagate trajectory **fused into one launch**
- K sized analytically from free VRAM; unit-flush cached at construction
- readout stays **on device** — only counts and projections cross the bus
- two packed splits chosen per formula; out-of-envelope formulas rejected, never silently degraded

Bottom line

Up to 5.7× end-to-end over the pre-WAVE engine — with **bit-identical model sets** at every step.

Negative results, kept because they cost weeks: classic **2-watched-literals on GPU is 1.8–2.6× slower** than NobleCount-style counting, on ∼5× the per-row state; 1-byte clause state, clause renumbering and multi-warp tiles all measured out.

Backup: the same suite on two machines — and why the ratio moved

family	targets	4080 laptop	**Colab L4**	wins (L4)
ITC'99 + ISCAS-89 seq, T=25	**rare**	36.5×	39.1×	17/18
ISCAS-85 combinational	**rare**	538×	615×	10/10
HWMCC'17, T=25	**rare**	156×	196×	7/7
EPFL combinational	**rare**	6.4×	9.4×	13/17
Sketch / `sk_` CNFs	—	46.2×	98.9×	27/29

Every ratio is higher on the L4 — and none of it is Gpap.

Paired on identical cells: Gpap 0.96× (same engine, slightly slower card) while **CMSGen runs at** 0.41× — a Xeon @ 2.2 GHz against an `i7-13800H`. CMSGen's own propagation *and* decision counters independently both give 0.413×.

Hence the rule: never quote a ratio without the CPU.

CMSGen is single-core, so the number is as much a property of the host as of the sampler. Otherwise a reviewer reads 98.9× and correctly calls it a rigged baseline. The same caution applies in advance to the Technion cluster runs.

Backup: two corrections we made to our own headline

One rule

A cell that measures the *instance* instead of the *sampler* does not belong in a geomean.

(i) Vacuous output pins. b02/b05/b07/s526 have constant outputs at $T{=}25$, so pinning them is a tautology and *any* sampler scores $\sim$100%. Including them read 47.7× on the sequential-rare column instead of 36.5× — an inflation of up to 1.9×, falling almost entirely on the column we tell people to quote.

(ii) Instances that need no search at all. A few cells emit samples with **zero decisions and zero implications**: after propagation the support is unconstrained, so "sampling" is a random-bit readout at 240M/s. Worse, picking the best encoder per instance *prefers* them, because they carry the family's highest ratio. Including them read EPFL 12.7× instead of 6.4×, and `sk_` 48.5× instead of 46.2×.

Both are now enforced in the table generator and re-checked by a verifier, so they cannot come back. The same generator produces the paper tables and these slides.

Backup: status, and what is open

Done

- engine: packed kernels, scalar + WAVE, `libgpap`, three language faces
- sampler: pipeline, portfolios, mining, ELS, GPU projection
- front-ends: `.bench`, AIGER, ABC, simulation
- evaluation: resumable sweep, 3-seed medians, fair CMSGen, 3 GPU rivals
- uniformity: 3-tier readout, Barbarik, exact-oracle fuzzing

Open — build

- **uniformity-aware config choice** — the one item that changes what a user gets
- per-instance encoder auto-pick
- state arena ($\sim$1.2$\times$)
- ROCm port (no kernel work)
- re-test the other applications

Open — measure

- **A100 rerun** — every number is K-bound on a 12 GB card
- multi-core CMSGen, QuickSampler
- break-even simulation throughput
- Barbarik $\times$10 / ScalBarbarik
- one single-session ablation

Backup: does the rival verdict survive a different GPU?

rival	protocol	GPAP / rival (geomean)		GPAP wins
		4080 laptop	**L4**	on L4
DEMOTIC	their ISCAS-85 pins	2.9×	2.5×	19/20
DEMOTIC	our **rare** pins	7,704×	2,427×	16/16
Recurrent	our **rare** pins	599×	422×	3/3
HTS	our CNFs, both encoders	—	2,042×	10/10

What transfers: the verdict.

Win counts are *identical* on both cards.

What does not: the magnitude.

Read these as orders of magnitude.

At rare pins the rival scores ≈0, so the ratio has a near-zero denominator. Honest: `c1355` loses on both cards.

Backup: numbers on hand

Engine

- vs pre-WAVE: b14 5.7×, b12 2.8×, b04 1.8×, b03 1.5×
- rates: b03 631k, b06 945k, b04 81k, b12 50k, b14 4.0k attempts/s
- C++ port: 1.013× geomean, identical model sets
- portability: 75/1273 CUDA lines, 18 driver calls, 29/29 specializations cross-compile

Encoder

- ABC shrink 2.2–7.8× clauses; at rare targets bimodal (unlocks b14, collapses b13)
- `fast` vs `full`: b15 encode 99 s → ≤4 s

Uniformity

- 600 instances, 10,800 runs, 0 unsound; controls 0/600 FP, 600/600 power
- $B \geq |\mathrm{sup}|$: 24/24 ok, TV 0.0392 (theory floor 0.0399)
- circuits 0.040 vs CMSGen 0.087; CNF 0.441 vs 0.272
- `soft`: Reject 5/5 in 156–936 samples, TV median 0.694

Baselines / method

- CMSGen emission = 5.3× its wall (373 MB on 27.sk)
- rare-target amplification 500–500,000×
- SM clock 615–1710 MHz while >90% busy; draw noise ±40%